DOC-17 | CONTROLLED SOURCE SYNCHRONIZATION

Cross-reference-only revision notice

v1.0-RC3 | 11 September 2026 | SOURCE-OF-TRUTH SYNCHRONIZATION ONLY

Control	Selected value
Revision	v1.0-RC2 -> **v1.0-RC3**
Semantic change	**NONE.** This revision changes source version/count/path references only.
Wave-2 semantic authorities	DOC-07 RC2 | DOC-08 RC2 | DOC-15 RC8 | DOC-16 RC5
Canonical HTTP companion	contracts/http/BLOG-DOC-15_openapi_v1.0-RC8.yaml
OpenAPI SHA-256	bf577a463b4c5d8fdceddc2087f53216426ef88d67d8e1a0d22ccf819751cd00
RC8 estate	113 operations | 86 paths | 221 schemas/helpers | 1,311 resolved internal refs | 0 unresolved refs 12 AUTH | 2 PUBLIC/PREVIEW | 63 SITE | 36 GLOBAL

DOCUMENT-SPECIFIC SYNCHRONIZATION

• Cross-reference update only: HTTP authority is DOC-15 v1.0-RC8 and discovery authority is DOC-16 v1.0-RC5.

• All inherited media/storage/image/CDN semantics, tables, capabilities, variants, ownership rules and acceptance criteria remain unchanged.

SUPERSESSION AND HISTORICAL CLASSIFICATION

The inherited body that follows is retained unchanged in substantive page content. Any inherited non-owning reference to DOC-15 RC7/RC6, the old 112/85/217/1,297 estate, 11 AUTH operations, DOC-16 RC4, or the RC7 YAML filename is classified **HISTORICAL / SUPERSEDED REFERENCE** and is not normative in this revision. All other inherited owned semantics remain in force.

Occurrence ledger: RC4: inherited printed pages 1; pre-RC8 HTTP upstream label: inherited printed pages 1.

IMPLEMENTATION IMPACT: NONE. The accepted NAKSHATRA_MASTER_WAVE2_HANDOFF application/database/API behavior is unchanged by this document synchronization.

BLOG-DOC-17

Media, Object Storage, Image Processing & CDN Specification

Developer execution authority for site-owned media lifecycle, upload validation, S3-compatible storage keying, originals, derivatives, thumbnails, responsive image delivery, WebP/AVIF direction, metadata, usage safety, replacement/deletion, CDN behavior and cross-site protection.

14	10	3	7
isolated site namespaces	frozen media capabilities	media persistence tables	frozen media API operations

CORE MEDIA RULE

A media asset is owned by exactly one site from upload through deletion. Storage infrastructure and CDN infrastructure may be shared, but object keys, database rows, usages, derivatives, public eligibility and cache invalidation never lose the originating SiteContext. No filename, media_id, CDN reachability or object-store location can turn a site-owned asset into global/shared tenant data.

Document control	Value
Status	REVIEW CANDIDATE - RC2 reconciliation complete; no known blocking contradiction remains within DOC-17 scope. Formal freeze requires explicit approval.
Version	1.0-RC2
Date	10 September 2026
Frozen upstream baseline	BLOG-DOC-01 through BLOG-DOC-16. Latest API authority: DOC-15 v1.0-RC6 + its OpenAPI companion. Latest discovery authority: DOC-16 v1.0-RC4.
Primary authority	Media ownership/lifecycle; upload type/size/decoder validation; S3-compatible key rules; accepted original; dimensions/orientation; deterministic focal/crop behavior; variants/thumbnails; WebP/AVIF; responsive delivery; metadata semantics; usage tracking; replacement/delete safety; public/private delivery; CDN/cache rules.
Explicit non-authority	New permissions/pages/endpoints/DB columns; queue/job payload and retry scheduler (PDF-18); credential/session/anti-malware implementation and secret handling (PDF-19); capacity/DR/retention topology (PDF-20/24); exhaustive QA program (PDF-21); repository/framework choice (PDF-23); site visual composition (PDFs 26-27).

DEVELOPER LOCK TARGET

A developer must not guess which files are accepted, how tenant ownership reaches object keys, whether originals are directly public, which derivative keys exist, whether images are upscaled, how thumbnails crop, which format is preferred, what happens when processing fails, when replacement affects existing references, what blocks deletion, when a CDN URL is public-eligible, or how cache revocation works. Those decisions are locked here for RC2 review; implementation must satisfy the reconciliation record and acceptance gates in Sections 31-35.

Table of Contents / Document Map

Section	Subject
1	Purpose, authority and conflict rule
2	Frozen upstream facts and one-source-of-truth map
3	Media architecture doctrine and identifiers
4	SiteContext, ownership and 14-site isolation
5	Media lifecycle and state model
6	Accepted v1 media class and upload contract
7	Upload validation pipeline
8	S3-compatible object-storage topology and keys
9	Original object rules and integrity
10	Dimensions, orientation and pixel safety
11	Focal point and deterministic crop rules
12	Canonical derivative/variant catalog
13	Image processing algorithm
14	WebP/AVIF generation direction
15	Thumbnail rules
16	Responsive image selection and frontend contract
17	Editorial metadata semantics
18	Media usage tracking and derived usage index
19	Reuse and relationship safety
20	Replacement semantics and atomicity
21	Deletion/tombstone/purge safety
22	Public eligibility, Admin/Preview delivery and privacy
23	CDN URL resolution and origin protection
24	CDN caching, invalidation and revocation
25	Admin API integration
26	Public PageData integration
27	SEO/Open Graph/structured-data integration
28	Security boundaries and unsafe-content controls
29	Failure, retry and recovery behavior
30	Operational signals and later-worker handoff
31	Cross-document consistency audit
32	Controlled reconciliation registry
33	Traceability matrix
34	Comprehensive developer acceptance criteria

Section	Subject
35	Freeze checklist and final developer lock
Appendix A	Variant registry
Appendix B	Storage/delivery key examples
Appendix C	Usage mapping
Appendix D	Failure matrix
Appendix E	One-page developer execution reference

1. Purpose, Authority and Conflict Rule

1.1 Purpose

BLOG-DOC-17 converts the frozen Media capability, persistence, Admin flow, public-data and frontend boundaries into one deterministic media delivery system. It owns the media-specific decisions intentionally deferred by DOC-08, DOC-11, DOC-13, DOC-14, DOC-15 and DOC-16 while leaving their existing identities, fields, routes and permissions unchanged.

1.2 Authority boundary

Decision	Authoritative source	DOC-17 treatment
Site ownership / RLS	DOC-04 / DOC-08	Consume exactly; never weaken or replace with CDN/object-key checks.
Permissions	DOC-02 / DOC-10	Consume exact media.view/upload/edit/replace/delete and composite field rules.
Media capabilities	DOC-03	Implement FC-MED-01..10; do not add media feature families.
Media tables/columns	DOC-08	Use media, media_variants, media_usages exactly; do not add hidden columns/tables.
Admin pages/states	DOC-06 / DOC-10	Support ADM-S-MEDIA-01/02 without adding Admin pages.
Public media object shape	DOC-13	Resolve PublicMedia/MediaVariant values; never add storage internals.
Frontend presentation	DOC-12 / DOC-14	Supply media semantics/URLs; site apps own aspect treatment and UI.
HTTP operations/schemas	DOC-15	Consume exact operations; any missing transport is flagged, not invented.
SEO image selection	DOC-16	Deliver the same-site selected media; do not redefine fallback priority.

1.3 Conflict rule

CONFLICT RULE

If implementation requires a new database field/table, new API operation/query/response property, new permission, new Admin/Public Page ID, a cross-site media pool, dynamic arbitrary transform API, new public image sitemap, or editable focal-point field that is not already frozen upstream, stop and reconcile the owning upstream authority. DOC-17 must not hide the change inside storage or CDN code.

2. Frozen Upstream Facts and One-Source-of-Truth Map

Source	Locked fact consumed by DOC-17
DOC-01	14 isolated public sites; one shared backend/data/infrastructure estate.
DOC-02	media.view, media.upload, media.edit, media.replace, media.delete; view prerequisite and composite image-SEO rules.
DOC-03	Exactly ten required Media capabilities FC-MED-01..10; metadata includes alt, caption/title, credit/source; search/filter, reuse, variants and CDN delivery.
DOC-04	Trusted SiteContext; site-owned derived systems retain site_id; missing context never becomes global.
DOC-05	No new public Page ID/resource page is created by media delivery.
DOC-06	ADM-S-MEDIA-01/02 only; variant/CDN effects are system effects, not pages.
DOC-07	Featured, body and OG media are same-site; ArticleBody MEDIA block has media_id + optional alt_override/caption.
DOC-08	media/media_variants/media_usages persistence; object keys site-namespaced; ACTIVE/DELETED media status; FORCE RLS.
DOC-09	Public page-route authority stays separate; CDN/media resource locators are not new Page IDs.
DOC-10	Media Library and Detail load site-owned assets; Media Detail includes variants/dimensions/usages; exact action permissions.
DOC-11	Missing optional media degrades safely; no raw object key; exact variant/CDN/focal behavior delegated here.
DOC-12	Contextual alt rules; missing media stays text-first; responsive media must not distort; site design owns layout.
DOC-13	PublicMedia and MediaVariant shapes; URLs are HTTPS and storage internals are forbidden.
DOC-14	14 site apps consume media contracts; site-to-site imports and all-site public media caches are prohibited.
DOC-15	Seven site-scoped Media HTTP operations; RC6 carries Media Library search/filter/pagination/sort, usage_count, current variants, title/source metadata and conditional replace-metadata authorization; storage internals remain excluded.
DOC-16	OG image fallback selection, no cross-site image fallback, no v1 image sitemap extension.

ONE SOURCE OF TRUTH

DOC-08 remains the physical record authority. DOC-15 remains HTTP authority. DOC-13 remains public object-shape authority. DOC-17 owns media processing/delivery semantics only. Tables in this PDF that repeat an upstream field do so for correlation, not as a second schema.

3. Media Architecture Doctrine and Identifiers

3.1 Processing/delivery pipeline

MEDIA17-ARCH-01

```
Admin upload/replace -> API15 site-scoped operation
  -> trusted SiteContext + permission check
  -> bounded stream + signature/decoder validation
  -> private S3-compatible origin object
  -> DB-TBL-013 media commit
  -> derivative generation under same site/media identity
  -> DB-TBL-014 media_variants rows
  -> DB-TBL-023 media_usages maintained from authoritative references
  -> public-eligibility resolver
  -> CDN public/protected delivery
  -> DOC-13 PublicMedia / MediaVariant
  -> one site-local frontend renderer
```

3.2 DOC-17 identifier families

Family	Format	Use
Rule	MED17-*	Normative media/storage/delivery rule.

Family	Format	Use
Validation	VAL17-*	Upload/replace/metadata validation.
Variant	VAR17-*	Stable derivative purpose/key.
Failure	FAIL17-*	Failure/recovery outcome.
Trace	TR17-*	Cross-document traceability.
Acceptance	QA17-*	Release-blocking assertion.
Gap	GAP17-*	Genuine upstream contradiction/missing transport; never silently patched.

3.3 Shared infrastructure is not shared ownership

- One S3-compatible service may physically host objects for all sites, but every canonical key starts inside a site namespace.
- One CDN distribution may serve all sites, but delivery lookup always validates the site namespace and media owner.
- One processing service may generate variants for all sites, but every work item retains the source site identity; exact queue payload/scheduler is later PDF-18.
- There is no cross-site Media Library, shared media record, global asset picker or fallback-by-filename behavior.

4. SiteContext, Ownership and 14-Site Isolation

MED17-ISO-01

```
assert ctx.site_id is trusted and resolved
assert media.site_id == ctx.site_id
assert object_key begins with canonical site namespace for ctx.site_id
assert each variant.site_id == media.site_id
assert each usage.site_id == media.site_id
assert referenced Post/Author/SiteSetting belongs to same site
if any check fails: non-disclose target; never retry without site scope
```

Boundary	Allowed	Prohibited
Upload	Create inside current ACTIVE site namespace.	Client-selected raw site_id or shared/global bucket prefix as ownership.
Picker/reuse	Return same-site ACTIVE media IDs only.	Search all media then filter in browser.
Variant job	Carry originating site_id + media_id.	Infer tenant from media_id alone when source context is known.
CDN lookup	Resolve site_key -> site_id -> same media owner.	Serve SITE-02 object through SITE-01 path due matching media_id/filename.
Missing media	Omit/degrade within same site.	Borrow an asset from another site.
Replace/delete	Mutate exactly one site-owned media record.	Move media to another site or rewrite foreign tenant references.

5. Media Lifecycle and State Model

DOC-08 freezes only ACTIVE and DELETED as durable media.status values. DOC-17 therefore does not add PROCESSING, FAILED, QUARANTINED or READY database states. Processing readiness is derived from validated object existence and derivative availability; worker execution state belongs later PDF-18.

Lifecycle moment	Durable DB state	Object/variant state	Delivery rule
Upload request streaming	No row yet	Temporary/staging only	Never public.

Lifecycle moment	Durable DB state	Object/variant state	Delivery rule
Validated original committed	ACTIVE	Canonical original stored privately; variants may be empty	Eligible for Admin use. Anonymous public delivery requires a verified sanitized native_webp baseline plus public-eligibility predicate.
Derivative generation	ACTIVE	Sanitized native_webp baseline first, then remaining fixed variants	Only verified sanitized derivative rows are public; canonical original never becomes anonymous fallback.
Metadata edit	ACTIVE	Binary unchanged	Binary remains; public PageData/cache metadata invalidation required later PDF-18.
Replacement commit	ACTIVE	New private original + verified native_webp are staged before atomic generation switch; remaining variants regenerate after	Same media_id/references; no public switch to an unsanitized or baseline-missing generation.
Delete accepted	DELETED	Derivative rows withdrawn; blobs become purge candidates	No new reference; all public/protected delivery denied.
Physical purge	DELETED/tombstone per DOC-08	Objects removed only under approved later security/retention policy	No resurrection from object storage.

NO HIDDEN STATUS

UI may show transient words such as Uploading or Processing as local/request state, but those labels are not persisted media.status and must not be used as authorization/public-eligibility truth.

6. Accepted v1 Media Class and Upload Contract

6.1 Accepted original MIME allowlist

MIME	Extensions accepted as hints	v1 status	Notes
image/jpeg	.jpg, .jpeg	ACCEPT	Single-frame raster.
image/png	.png	ACCEPT	Single-frame raster; alpha allowed.
image/webp	.webp	ACCEPT	Static/single-frame only.
image/avif	.avif	ACCEPT	Static/single-frame only.
image/gif	.gif	REJECT	Animation/multi-frame path not approved in v1.
image/svg+xml	.svg	REJECT	Active/vector document processing not approved in v1.
Other image/video/audio/PDF	any	REJECT	Would expand the frozen v1 media processing contract.

6.2 Hard resource limits

Limit	Locked v1 value	Validation rule
Original upload bytes	20 MiB maximum	Streaming byte counter is authoritative; Content-Length is only an early rejection hint.
Width	1..12,000 px	Decoded display width after orientation interpretation.
Height	1..12,000 px	Decoded display height after orientation interpretation.
Pixel area	80,000,000 px maximum	width * height must not exceed 80 MP.
Frames/pages	Exactly 1	Animated/multi-frame/multi-page image rejected.
Filename	Editorial label only	Never used as object key, path authority or MIME authority.

6.3 Why the v1 class is intentionally narrow

The frozen product describes an image-heavy Media Library and explicitly delegates image processing/variants to DOC-17. Accepting SVG, GIF animation, video, audio or arbitrary documents would require new processing, security, renderer and possibly public-object rules. DOC-17 therefore does not smuggle those capabilities into a generic MIME field.

7. Upload Validation Pipeline

1. Resolve trusted SiteContext and enforce media.view + media.upload before durable media work.
2. Apply streaming 20 MiB hard limit; never buffer an unbounded request in memory.
3. Sanitize original_filename as metadata only: remove control characters/path separators from display value; do not derive ownership or key from it.
4. Detect actual file signature and decoder type. Declared multipart MIME/extension must agree with the decoded allowlisted image class.
5. Decode under bounded width/height/pixel/frame limits. Decoder failure, decompression bomb behavior, unsupported color model or multi-frame content fails validation.
6. Read intrinsic dimensions and orientation. Dimensions persisted in DOC-08 describe the display orientation used by the processing pipeline.
7. Compute SHA-256 over the canonical accepted original bytes that will be stored. The checksum stays internal.
8. Write the canonical original to a site-namespaced PRIVATE origin key using create-only/conditional semantics. Uploaded EXIF/comment metadata may remain in this private preservation object; it is never an anonymous public representation.
9. Create the media row under the same site transaction/context. Do not create a durable row that points to a missing original.
10. Generate and verify native_webp as the mandatory sanitized full-dimension public baseline. Remaining fixed derivatives may continue asynchronously under the same site/media identity; exact queue payload/retry cadence belongs PDF-18.
11. Return DOC-15 MediaResponse. A 201 response may precede completion of non-baseline variants, but ordinary PublicMedia is emitted only after native_webp exists and MED17-PUBLIC-01 passes. Canonical original bytes are never the public fallback.

Validation ID	Condition	DOC-15 outward class
VAL17-UP-01	No trusted ACTIVE SiteContext / wrong-site request	Use existing 403/404 non-disclosure semantics; no storage write.
VAL17-UP-02	Byte limit exceeded	422 validation error; 413 is not invented because DOC-15 did not define it.
VAL17-UP-03	Unsupported signature/MIME or mismatch	422.
VAL17-UP-04	Decode/dimension/pixel/frame violation	422.
VAL17-UP-05	Storage dependency unavailable before commit	503; no durable media row.
VAL17-UP-06	Idempotency same key/different fingerprint	409 under DOC-15.
VAL17-UP-07	Duplicate checksum in same/different site	Not an error by itself; no automatic cross-record/cross-site deduplication.

NO CROSS-SITE DEDUP

A checksum match never makes two site-owned media records share ownership. Physical backend deduplication, if ever introduced below the logical object layer, must be invisible to keys, authorization, purge and tenant semantics and cannot be assumed by this v1 contract.

8. S3-Compatible Object-Storage Topology and Key Contract

8.1 Logical storage classes

Class	Access	Purpose
MEDIA_ORIGIN	Private to application/processing/CDN origin identity	Canonical accepted original for each media generation. Never an anonymous object-store URL.
MEDIA_DERIVATIVE	Private origin behind delivery/CDN layer	Generated WebP/AVIF responsive and thumbnail objects.
MEDIA_STAGING	Private, temporary	Upload/replace write-before-commit staging. Never referenced by DB/PublicMedia.

Actual provider bucket names, regions, credentials, replication and deployment topology are configuration/deployment concerns. The logical classes and key grammar below are DOC-17 authority.

8.2 Canonical key grammar

MED17-KEY-01

```
ORIGINAL KEY
sites/{site_id}/media/{media_id}/g/{checksum_sha256}/original.{canonical_ext}

VARIANT KEY
sites/{site_id}/media/{media_id}/g/{checksum_sha256}/variants/{variant_key}.{variant_ext}

STAGING KEY (never persisted)
_staging/sites/{site_id}/{request_id}/{opaque_upload_id}
```

- site_id and media_id are server-derived identifiers; client filenames never enter the authority portion of a key.
- checksum_sha256 is allowed in the private object key but is forbidden from DOC-13 public objects and public URL construction.
- Every persisted media.object_key and media_variants.object_key remains unique inside the site as already enforced by DOC-08.
- Replacement creates a new generation path; no object is overwritten in place before validation succeeds.
- Path traversal tokens, raw slashes from filenames and user-supplied key fragments are never accepted.

8.3 Object metadata/tags

Storage metadata	Required behavior
Content-Type	Exact validated stored MIME.
Cache-Control on origin	Origin objects are private; public cache policy is applied by delivery/CDN layer, not trusted from uploader metadata.
Site/media tags	May mirror site_id/media_id for operations, but tags are not authorization.
Filename	May be stored as safe Content-Disposition metadata for Admin download only if later transport allows; never key authority.
User metadata	Arbitrary x-amz-meta-* passthrough from client is prohibited.

9. Original Object Rules and Integrity

- The canonical original is the accepted single-frame raster payload in its validated input format. It preserves source pixel dimensions and may preserve uploader/source metadata for private fidelity; DOC-17 does not resize/re-encode this preservation object.
- The canonical original is PRIVATE ONLY. It is never used as PublicMedia.url, never appears in public variants[], and is never an anonymous CDN fallback. This prevents embedded GPS/EXIF/comments from becoming public by accident.

- Every anonymously public representation is a DOC-17 generated derivative that strips GPS, EXIF, comments and other nonessential/application metadata after orientation is baked into pixels. native_webp is the mandatory sanitized full-dimension public baseline.
- DOC-08 media.mime_type, byte_size, width_px, height_px, checksum_sha256 and object_key describe the canonical stored original, not a derivative. PublicMedia.mime_type/url may therefore describe native_webp instead of media.mime_type/object_key.
- The origin object is immutable for one generation. Replacement writes a new generation original and a verified native_webp before switching the current generation; it never mutates the existing object in place.
- Original object HEAD/GET is never anonymous from S3-compatible origin or CDN. Admin/Preview may receive a protected no-store representation under their existing authority; public delivery resolves only sanitized derivative rows.
- If DB commit fails after an origin write, that object is orphan-staging cleanup work and cannot be returned as MediaResource/PublicMedia.

9.1 Integrity checks

Check	When	Failure
SHA-256	After canonical accepted bytes are finalized and on integrity diagnostics	Mismatch -> do not serve as current; recover from valid generation/backup according to later operations policy.
Byte size	Upload and object HEAD verification	Mismatch -> processing failure; no derivative trust.
MIME	Signature/decoder + stored Content-Type	Mismatch -> fail closed.
Dimensions	Decoder result vs persisted values	Mismatch -> recompute only through controlled repair path; do not let CDN metadata redefine DB truth.

10. Dimensions, Orientation and Pixel Safety

MED17-DIM-01

```
W,H = decoder display dimensions after applying orientation semantics
require 1 <= W <= 12000
require 1 <= H <= 12000
require W * H <= 80000000
persist width_px=W, height_px=H
for uncropped derivative target T:
    if W <= T: omit that width variant (no upscale)
    else out_w=T; out_h=max(1, round(H*T/W))
```

- Aspect ratio is preserved for all uncropped responsive variants.
- Thumbnails are the only v1 catalog entries that crop; Section 11/15 owns their crop.
- A site frontend may visually crop an image in CSS, but it must not request an undocumented transform or modify the underlying media semantics.
- Intrinsic dimensions returned through DOC-13 allow frontends to reserve layout space and reduce cumulative layout shift; exact site layout remains DOC-14/26-27 authority.

11. Focal Point and Deterministic Crop Rules

11.1 Frozen v1 focal source

NO EDITABLE FOCAL FIELD IN FROZEN UPSTREAM

DOC-08 and DOC-15 contain no focal-point persistence/transport field. DOC-17 therefore does not invent one. The v1 effective focal point is exactly the normalized center (fx=0.5, fy=0.5) for generated crop variants. This fulfills deterministic crop behavior without silently changing frozen schemas. A future editor-controlled focal point requires a controlled DOC-08 + DOC-15 revision before DOC-17 can consume it.

11.2 Crop algorithm

MED17-CROP-01

```
input: source W,H; target tw,th; focal fx=0.5, fy=0.5 in v1
r = tw / th
crop_w = min(W, H * r)
crop_h = crop_w / r
left = clamp(fx*W - crop_w/2, 0, W-crop_w)
top  = clamp(fy*H - crop_h/2, 0, H-crop_h)
crop = source[left:left+crop_w, top:top+crop_h]
resize crop to tw x th only if both crop dimensions >= target dimensions
otherwise omit the target thumbnail; never upscale
```

11.3 Frontend focal boundary

No focal coordinates are emitted in PublicMedia. Site frontends consume already-generated crop variants or perform their own CSS object-fit presentation. They may not infer a different editorial focal point from filenames, face detection or another site asset.

12. Canonical Derivative / Variant Catalog

DOC-08 persists a stable variant_key and DOC-13 exposes that key publicly. DOC-17 therefore freezes the v1 catalog below. Variants are generated only when source dimensions permit without upscaling.

ID	variant_key	Geometry	MIME	Rule
VAR17-TH-256-AVIF	thumb256_avif	256x256 crop	AVIF	Optional optimization; center focal crop.
VAR17-TH-256-WEBP	thumb256_webp	256x256 crop	WebP	Required thumbnail baseline when source can satisfy.
VAR17-TH-512-AVIF	thumb512_avif	512x512 crop	AVIF	Optional optimization.
VAR17-TH-512-WEBP	thumb512_webp	512x512 crop	WebP	Required when source can satisfy.
VAR17-W320-A	w320_avif	max width 320	AVIF	Uncropped responsive.
VAR17-W320-W	w320_webp	max width 320	WebP	Uncropped responsive.
VAR17-W640-A	w640_avif	max width 640	AVIF	Uncropped responsive.
VAR17-W640-W	w640_webp	max width 640	WebP	Uncropped responsive.
VAR17-W960-A	w960_avif	max width 960	AVIF	Uncropped responsive.
VAR17-W960-W	w960_webp	max width 960	WebP	Uncropped responsive.
VAR17-W1280-A	w1280_avif	max width 1280	AVIF	Uncropped responsive.
VAR17-W1280-W	w1280_webp	max width 1280	WebP	Uncropped responsive.
VAR17-W1600-A	w1600_avif	max width 1600	AVIF	Uncropped responsive.
VAR17-W1600-W	w1600_webp	max width 1600	WebP	Uncropped responsive.
VAR17-NATIVE-A	native_avif	original dimensions	AVIF	No crop/upscale; modern-format full-size derivative.
VAR17-NATIVE-W	native_webp	original dimensions	WebP	No crop/upscale; baseline full-size derivative.

12.1 Catalog ordering in DOC-13 variants[]

Emit only available/current variants, in this deterministic order: thumb256_avif, thumb256_webp, thumb512_avif, thumb512_webp, w320_avif, w320_webp, w640_avif, w640_webp, w960_avif, w960_webp, w1280_avif, w1280_webp, w1600_avif, w1600_webp, native_avif, native_webp. The frontend preserves server order but selects by variant_key/type rather than by array position.

12.2 No dynamic transform API

- No arbitrary width/height/quality/crop query parameters are accepted from public clients.
- No runtime URL-based image transformation service is required for v1.

- A new derivative purpose requires a controlled DOC-17 catalog revision; it is not created by a site frontend or Admin input.

13. Image Processing Algorithm

1. Open only the current site-owned ACTIVE media generation under trusted SiteContext.
2. Decode the private canonical original with limits from Section 6/10 and interpret orientation.
3. Convert derivative working pixels to sRGB while preserving alpha where source requires it.
4. Generate native_webp FIRST at oriented source dimensions, without crop/upscale, strip all nonessential metadata, verify object MIME/bytes/dimensions and persist the variant row. This is the public baseline.
5. Generate the remaining required WebP entries next. Failure of a non-baseline WebP derivative may degrade responsive choice but cannot authorize exposure of the original.
6. Generate AVIF entries as preferred enhancement. AVIF failure must not remove an already-valid sanitized WebP baseline.
7. For w* and native* entries preserve source aspect ratio and never upscale.
8. For thumb* entries apply Section 11 center-focal square crop and never upscale.
9. Strip GPS/EXIF/comment/application metadata from EVERY derivative; retain only safe color/orientation information already baked into pixels.
10. Write derivative bytes to a generation-specific private object key before inserting/updating media_variants.
11. A media_variants row is created only after the corresponding object write succeeds and its byte size/dimensions/MIME are verified.
12. Never expose a partially written object or a variant row whose object is missing.
13. On retry, the same site/media/generation/variant_key is idempotent: verified identical output may be reused; invalid partial objects are replaced before DB row commit.

13.1 Variant identity across replacement

variant_key is the stable semantic key. variant_id is a derived persistence-row identity and need not remain stable when a replacement rebuilds the derivative set. Public clients depend on variant_key, URL, MIME and dimensions, not variant_id.

14. WebP / AVIF Generation Direction

Concern	WebP	AVIF	Private canonical original
Role	Required sanitized baseline; native_webp is mandatory before anonymous public delivery.	Preferred additional compression/modern source.	Preservation/source object only; never anonymous public fallback.
Photo/default quality	quality 82	quality 60	No DOC-17 public optimization role.
PNG/alpha source	Lossless WebP for PNG/graphic source; alpha preserved.	quality 70; alpha preserved.	Private original PNG remains preservation object only.
Metadata	Strip GPS/EXIF/comments/nonessential application metadata from every public derivative.	Same stripping rule.	May retain accepted source metadata because access is private/protected only.
Failure	If native_webp fails, PublicMedia is unavailable until repaired; other WebP failures degrade choices only.	Do not block public rendering when sanitized WebP baseline is valid.	Private original can support repair/Admin/Preview, but must never be substituted into anonymous public delivery.

FORMAT SELECTION

Site frontends must prefer AVIF when a matching AVIF source exists, then WebP, then PublicMedia.url. PublicMedia.url itself resolves to the sanitized native_webp baseline, never to canonical original bytes. They do not rewrite an AVIF URL to WebP or construct a storage key.

15. Thumbnail Rules

- thumb256_* and thumb512_* are square editorial/card preview derivatives, not a new media object.
- Thumbnail crop uses exactly MED17-CROP-01 and the v1 center focal point.
- Generate thumb256 only when the crop window can produce at least 256x256 without upscaling; generate thumb512 only when it can produce at least 512x512.
- Thumbnails contain no baked text, watermark, site logo, border, rounded corner or theme treatment; those are frontend presentation concerns.
- A Media Library may use thumbnail variants for grid efficiency, while Asset Detail may choose a larger uncropped variant or a protected no-store original representation.
- Missing thumbnail does not mean media is missing. The renderer falls back to an available sanitized uncropped derivative within the same media object; anonymous public rendering never falls back to canonical original bytes.

16. Responsive Image Selection and Frontend Contract

16.1 Server responsibilities

- DOC-13 PublicMedia.url is the resolved same-site sanitized native_webp baseline URL. PublicMedia.mime_type is image/webp and width/height describe the oriented full-dimension native_webp source.
- PublicMedia.variants contains only current verified derivative rows in the fixed DOC-17 order.
- Every variant URL is already resolved; the frontend never reconstructs an object key or CDN path.
- No cross-site or private-original fallback is returned when a derivative is unavailable.

16.2 Site-local frontend responsibilities

MED17-RSP-01

```
preferred uncropped picture:
  AVIF srcset = available w320..w1600/native AVIF variants
  WebP srcset = available w320..w1600/native WebP variants
  img.src = PublicMedia.url
  img.width/height = known intrinsic dimensions when available
  img.alt = DOC-12 contextual alt rule
  sizes = site-local responsive layout decision

card square preview:
  select thumb256/thumb512 by viewport/DPR when present
  otherwise use an uncropped same-media source; CSS may crop visually
```

- Do not use variant array order as a substitute for matching variant_key/MIME.
- Do not reorder semantic article/card lists based on image dimensions.
- Do not stretch images; preserve aspect ratio unless a site-local presentation intentionally crops inside a bounded visual frame.
- Lazy/eager loading policy is site-local/performance authority; DOC-17 only supplies correct sources.

17. Editorial Metadata Semantics

Field	Locked semantic
alt_text	Default approved alternative text for contexts where media contributes meaning. Blank/whitespace Media Library input normalizes to absent; contextual decorative empty-alt remains renderer logic under DOC-12.
title	Editorial media title persisted by DOC-08; not the article H1, not automatically emitted as HTML title/alt.
caption	Default human-visible caption when the consuming context chooses to render one. Body MEDIA block caption may override for that block.
credit_text	Public credit/source display text when provided and projected. DOC-15 transports this as credit where available.
source_url	Editorial source reference persisted by DOC-08. It is not fetched/imported automatically and is not part of DOC-13 PublicMedia.
original_filename	Editorial upload provenance only; never public URL/key/ownership authority.
width_px/height_px	System-derived intrinsic display dimensions; not user-editable metadata.

Field	Locked semantic
mime_type/byte_size/ checksum/object_key	System/infrastructure fields; checksum/object_key never public.

17.1 Context override hierarchy

Context	Alt	Caption / credit
Article body MEDIA	block.alt_override when nonblank -> media.alt_text -> context-safe empty/omission behavior	block.caption when supplied -> media.caption; credit_text remains media-level when projected.
Article featured	media.alt_text if informative; card may intentionally emit empty alt when redundant with linked title	media.caption/credit only when component contract renders them.
Author avatar	Empty alt when redundant with adjacent author name; otherwise media.alt_text	Normally no caption unless owning component contract allows.
OG/structured data	DOC-16 chooses media; alt does not affect canonical image URL selection	No fabricated credit in JSON-LD unless approved output contract uses it.

17.2 Metadata transport reconciliation

DOC-15 RC6 reconciles the existing media metadata transport: MediaUploadInput, MediaUpdateInput and MediaResource carry title and source_url in addition to alt_text, caption and credit. PATCH metadata uses media.view + media.edit. Replace remains media.view + media.replace for binary replacement, and supplying any metadata field during replace additionally requires media.edit; failure is atomic.

18. Media Usage Tracking and Derived Usage Index

DB-TBL-023 media_usages is a derived index, not the referential source of truth. It is rebuilt from current authoritative fields/body content and used for visibility and delete safety. A stale usage index must never authorize destructive deletion.

DOC-08 usage_kind	Authoritative source	DOC-15 coarse resource_type
POST_FEATURED	posts.featured_media_id	POST
POST_BODY	Validated ArticleBody MEDIA block media_id; block_id retained internally	POST
POST_OG	posts.og_image_media_id	POST
AUTHOR_PHOTO	author_site_profiles.photo_media_id	AUTHOR
SITE_LOGO	site_settings.logo_media_id	SITE_SETTING
SITE_FAVICON	site_settings.favicon_media_id	SITE_SETTING
SITE_DEFAULT_SOCIAL	site_settings.default_social_image_media_id	SITE_SETTING

18.1 Update algorithm

MED17-USAGE-01

```
on any authoritative media-reference commit:
  derive exact current usages for affected owner inside same site
  replace/reconcile media_usages rows transactionally with owner mutation
  never derive a usage from another site
on delete preflight:
  read usage index for fast impact list
  re-check authoritative current references before destructive commit
  if any blocking reference remains: reject delete
```

18.2 Usage response privacy

- The usage endpoint requires media.view. Opening a referenced Post additionally requires posts.view under DOC-10.
- If the caller lacks read authority for a referenced target, the response label must not leak protected target content; a generic usage label may be returned while preserving the fact that deletion is blocked.
- Multiple usage kinds in one Post may appear as separate rows/items when needed to explain impact. The API does not grant mutation permission on the consuming resource.

19. Reuse and Relationship Safety

1. Media picker starts from trusted SiteContext and media.view.
2. Only ACTIVE media rows from the same site are candidates.
3. The caller must also hold the target mutation permission (for existing Post reuse: posts.view + posts.edit).
4. On assignment, the target service revalidates media.site_id against target.site_id; UI selection is not authority.
5. Usage index is updated from the committed reference.
6. No binary is duplicated merely because the same site reuses one media_id in multiple Posts.
7. Cross-site reuse is prohibited. Re-uploading equivalent bytes in another site creates a distinct site-owned media record/object namespace.

20. Replacement Semantics and Atomicity

20.1 Replacement invariant

> **SAME ID, NEW BINARY GENERATION**
>
> A successful replace keeps the existing media_id and all valid same-site references. It changes the current original binary generation and rebuilds variants. It is not a delete+new media operation and it never changes site_id.

20.2 Safe replace algorithm

1. Authorize media.view + media.replace and resolve target inside current site. If the request also supplies alt_text/title/caption/credit/source_url, additionally require media.edit before any binary or metadata switch; authorization failure rejects the whole request.
2. Re-read current active usages to show impact; usage does not prohibit replace, but the operation affects every current usage of that media_id.
3. Validate the new file with the same Section 6/7 rules as upload.
4. Write the new generation canonical original under a new immutable PRIVATE object key. Do not alter the current media row yet.
5. Generate, sanitize and verify the new generation native_webp derivative before the durable switch. If native_webp cannot be produced, abort replacement and keep the old generation current/publicly deliverable.
6. Within one serialized transaction, verify the target is still ACTIVE and same-site; withdraw old current variant rows, update original object_key/checksum/MIME/bytes/dimensions and updated_at, insert the verified new native_webp variant row, and apply only authorized supplied metadata fields. Omitted metadata preserves current values.
7. Commit. All consuming references still point to the same media_id, and the new generation already has a safe public baseline.
8. Generate the remaining fixed WebP/AVIF derivatives for the new generation; partial non-baseline availability is permitted.
9. Purge/revoke CDN cache for the old generation/current media namespace. Old objects become cleanup candidates under later retention/security policy.
10. If any validation/origin-write/native_webp/precommit check fails, the old media generation and its sanitized public baseline remain authoritative and deliverable.

20.3 Concurrent replace/delete behavior

DOC-15 exposes no media expected_version token. The service must still serialize destructive/current-generation switches with a row lock or equivalent conditional transaction so two replace/delete requests cannot interleave into a DB/object mismatch. This is internal atomicity and does not change the DOC-15 concurrency contract.

21. Deletion, Tombstone and Physical Purge Safety

21.1 Delete eligibility

MED17-DEL-01

```
authorize media.view + media.delete
resolve ACTIVE same-site media
usage_hint = media_usages for media
blocking_refs = authoritative current references revalidated in transaction
if blocking_refs != empty: reject with existing DOC15 conflict/error semantics
else:
  set media.status=DELETED; deleted_at/by = current actor/time
  withdraw media_variants rows from current delivery catalog
  revoke public/protected delivery and purge CDN namespace
  commit tombstone
physical object purge occurs only under approved later retention/security policy
```

21.2 Blocking vs non-blocking references

Reference class	Delete effect	Reason
Current Post featured/body/OG	BLOCK	Would break current content/editable relationship.
Current Author photo	BLOCK	Would break current profile relationship.
Site logo/favicon/default social	BLOCK	Would break current site setting.
Historical Post revision snapshot	NON-BLOCKING by itself	DOC-08 allows historical snapshots to reference tombstoned media; restore revalidates ACTIVE media before reuse.
Orphan media with no current reference	ALLOW after recheck	Tombstone + delivery revocation.

21.3 No resurrection by CDN

- DELETED media is never emitted into new PublicMedia/Admin selection results.
- CDN cache purge is mandatory on deletion and on loss of all public eligibility.
- An old cached delivery URL must not be treated as proof the media is still ACTIVE.
- Physical blob retention does not mean public availability.

22. Public Eligibility, Admin/Preview Delivery and Privacy

22.1 Publicly deliverable predicate

MED17-PUBLIC-01

```
is_publicly_deliverable(media, site):
  media.status == ACTIVE
  AND media.site_id == site.site_id
  AND site.status == ACTIVE for ordinary public delivery
  AND media is selected by a currently valid same-site public projection/reference
      (PUBLISHED Post featured/body/OG, public-eligible Author photo,
       site logo/favicon/default social media as applicable)
  AND requested delivery site namespace matches owner
  AND a verified current-generation native_webp variant exists for anonymous public delivery
```

The predicate is semantic authority. A derived eligibility cache may accelerate it later, but a cache is not the source of truth and must remain site-keyed under PDF-18.

22.2 Delivery classes

Class	Audience	Cache	Rule
PUBLIC	Anonymous visitor/SEO crawler	Public CDN cache allowed	Only media passing MED17-PUBLIC-01; PublicMedia.url resolves to sanitized native_webp, never the canonical original.
PREVIEW	Valid private Preview authority	private, no-store	May render otherwise nonpublic same-site ACTIVE media; a protected original representation may be used, but canonical original is never made anonymous/public. Credential/session mechanics remain PDF-19.
ADMIN	Authenticated authorized Admin	private, no-store	Media Library/Detail preview may resolve same-site ACTIVE media regardless of public usage; never a public signed credential.
DELETED / wrong-site	Nobody	no-store error	No bytes and no tenant/object metadata disclosure.

22.3 Public URL does not imply global ownership

A CDN URL can be globally reachable on the Internet while the underlying media record remains site-owned. Ownership is established by trusted site identity and persistence relationships, never by who can technically fetch a public image URL.

23. CDN URL Resolution and Origin Protection

23.1 Logical delivery locator

MED17-CDN-URL-01

```
public baseline (PublicMedia.url):
  {MEDIA_CDN_ORIGIN}/m/{site_key}/{media_id}/native_webp.webp?v={updated_at_epoch_ms}
public variant:
  {MEDIA_CDN_ORIGIN}/m/{site_key}/{media_id}/{variant_key}.{ext}?v={updated_at_epoch_ms}
protected Admin/Preview original delivery:
  implementation-specific protected media resource under the configured delivery origin;
  it must carry site_key + media_id semantics and use session/Preview authority, not an object-store key.
```

- MEDIA_CDN_ORIGIN is deployment configuration, not a site-owned database field and not the canonical website domain.
- The public URL never contains site_id, object_key, bucket, checksum, origin credential or private signed token.
- site_key is public identity only. The delivery service still resolves it to the owning internal site and verifies media ownership.
- The v query is based on media.updated_at and exists only as a cache revision hint. It is not authorization. Metadata-only edits may change v; correctness is preferred over binary-cache reuse.
- Public PageData returns fully resolved HTTPS URLs. A site frontend does not concatenate object-store or CDN paths itself for public media.
- Media delivery paths are resource delivery, not new BLOG-DOC-05 public Page IDs or editable route families.

23.2 Origin access

- S3-compatible origin buckets are not public; the canonical original is never anonymously delivered through the CDN either.
- CDN/origin service uses server-side identity to fetch only the object key resolved from the validated media/variant row.
- A requested site_key/media_id/variant_key tuple that does not resolve in one site returns a neutral not-found outcome without exposing the correct tenant.
- The origin must set/verify Content-Type from stored validated metadata and send X-Content-Type-Options: nosniff where the delivery stack supports it.

24. CDN Caching, Invalidation and Revocation

24.1 Cache policy

Response	Cache-Control direction	Notes
Public sanitized baseline/variant	public, max-age=31536000, immutable	Only generated metadata-sanitized derivatives; URL carries v revision and CDN query-string cache key includes v.
Preview media	private, no-store	Never populate shared public CDN cache.
Admin media preview	private, no-store	Never populate shared public CDN cache.
Wrong-site/not found/deleted/private through public path	no-store	Do not create a long-lived negative cache that can cross lifecycle/site changes.

24.2 Mandatory invalidation events

Event	Binary CDN action	PageData/cache handoff
Replace	Purge/revoke all previously current URLs for media_id; new v resolves new generation.	Affected public projections must be invalidated/rebuilt later under PDF-18.
Delete	Purge all media URLs and deny origin delivery.	Affected pages cannot retain now-invalid media; deletion is blocked while current refs exist.
Public eligibility revoked (e.g. last public usage disappears)	Purge public cache; subsequent anonymous fetch denied.	Derived public cache updates under PDF-18.
Metadata-only alt/title/caption/credit/source	Binary derivative purge not required for content bytes; v may change because updated_at changes.	Public PageData carrying metadata must refresh.
Variant regeneration same binary	Purge only affected variant path if an existing variant row/object is replaced.	Variant array refresh if availability changed.

24.3 Cache isolation

- CDN key includes site_key and media_id; there is no cache entry keyed by media_id alone.
- Any application-side cache of public eligibility/variant catalog is keyed by internal site identity as required by DOC-04/DOC-14.
- A purge request always carries/derives the site identity; global wildcard purge is not the normal media mutation path.
- Exact distributed cache implementation and worker event payload are PDF-18; DOC-17 owns the invalidation condition and required outcome.

25. Admin API Integration

25.1 Frozen DOC-15 media operations

Operation	Method/path	Success	Permission	DOC-17 semantic handoff
API15-S-060	GET /api/v1/admin/sites/{site_key}/media	200	media.view	RC6: same-site search, MIME + usage filter, page/page_size and deterministic sort; MediaResource items include usage_count + variants.
API15-S-061	POST .../media	201	media.view + media.upload	Multipart MediaUploadInput includes alt_text/title/caption/credit/source_url; apply Sections 6-9.
API15-S-062	GET .../media/{media_id}	200	media.view	RC6 MediaResource carries current verified variants/dimensions plus usage_count.

Operation	Method/path	Success	Permission	DOC-17 semantic handoff
API15-S-063	PATCH .../media/{media_id}	200	media.view + media.edit	Update alt_text/title/caption/credit/source_url only; no binary replacement.
API15-S-064	POST .../media/{media_id}/replace	200	media.view + media.replace; + media.edit when metadata supplied	Safe replacement Section 20; conditional metadata authorization is atomic.
API15-S-065	DELETE .../media/{media_id}	204	media.view + media.delete	Usage-revalidated tombstone Section 21.
API15-S-066	GET .../media/{media_id}/usage	200	media.view	Derived same-site usage list; opening target needs target view permission.

25.2 Existing schema mapping

DOC-15 RC6 schema field	Persistence/source	DOC-17 rule
filename	media.original_filename	Editorial only; sanitized display value.
mime_type	media.mime_type	Validated PRIVATE original MIME.
size_bytes	media.byte_size	Stored private original bytes.
width / height	media.width_px / height_px	Decoded intrinsic display dimensions.
alt_text / title / caption / credit / source_url	media metadata columns	Editable transport restored in RC6; source_url remains non-public by DOC-13.
usage_count	derived current usage set	Required nonnegative list/detail indicator; avoids per-item usage lookup merely to render used/unused state.
variants[]	media_variants current verified rows	Admin detail/list transport; byte_size + variant_key/MIME/dimensions, never object_key/checksum.
status	media.status	Only ACTIVE/DELETED; no processing status invented.
public_url	Resolved delivery layer	When present, resolves sanitized native_webp/current safe public baseline; never private original or signed credential.

25.3 Idempotency and errors

- Upload, metadata PATCH, replace and delete preserve the DOC-15 idempotency contract; DOC-17 does not change its 24-hour replay/fingerprint semantics. API15-S-060 pagination/filtering uses the exact RC6 query contract and remains server-side within one SiteContext.
- Type/size/dimension/decoder errors use 422 because DOC-15 did not define 413/415 for these operations.
- Usage conflict/delete-blocked or idempotency fingerprint conflict uses the existing DOC-15 conflict family; no new success status is invented.
- Provider/storage unavailability uses the existing 503 family.

26. Public PageData Integration

26.1 PublicMedia construction

MED17-PROJECT-01

```
input: current site + one approved media reference
resolve ACTIVE media in same site
if ordinary public: require MED17-PUBLIC-01
require verified current native_webp; resolve its sanitized CDN URL as PublicMedia.url
load current media_variants for same site/media/generation
filter to verified existing objects
sort by DOC17 fixed variant order
```

```
project only: media_id,url,mime_type,width?,height?,alt_text?,caption?,credit_text?,variants[]
never project: site_id,object_key,bucket,checksum,source_url,usage rows,origin credentials
```

26.2 Missing-media behavior

Context	DOC-17 output	Frontend behavior inherited
Optional featured/card/avatar/site image	Omit optional PublicMedia when it cannot be safely resolved.	Keep text/content usable; no broken path.
ArticleBody MEDIA block	resolution_state=MISSING and omit media object.	Render CMP12-MISSING-MEDIA; do not borrow cross-site.
Preview	Same-site protected delivery only; no public-cache substitution.	Remain private/partial.
Required primary media does not exist in current contract	Do not invent a required-image publish blocker.	Missing optional media does not invalidate otherwise valid Article.

26.3 Variant array contract

- variants is always present as a required array in PublicMedia. For any emitted ordinary-public PublicMedia, native_webp exists and is represented by PublicMedia.url; variants[] contains the verified current derivative catalog and may include native_webp as its canonical variant row.
- Only verified current-generation variants are projected.
- Public MediaVariant never exposes byte_size, object_key, variant_id or checksum.
- Variant URL/MIME/width/height are derived from the DB-TBL-014 row and delivery resolver.

27. SEO, Open Graph and Structured-Data Integration

SEO concern	Authority	DOC-17 responsibility
OG image choice	DOC-16	Deliver the media chosen by post OG -> featured -> site default fallback; do not choose a different asset.
Same-site image	DOC-04/16	Re-resolve chosen media in owning SiteContext and deny cross-site.
OG/JSON-LD URL	DOC-16 + DOC-13	Return only public-safe resolved CDN URL; no origin/object key.
Alt	DOC-12/16	Provide authored PublicMedia.alt_text; never fabricate for SEO.
Image sitemap	DOC-16	Not required in v1; DOC-17 does not add one.
External canonical	DOC-16	Does not permit borrowing external/cross-site image as the media object.

28. Security Boundaries and Unsafe-Content Controls

DOC-17 freezes media-domain safety controls that are inseparable from image ingestion/delivery. Authentication/session token formats, secret storage, anti-CSRF/CORS policy, malware-engine selection and broader platform security remain later PDF-19 authority.

Threat/control	DOC-17 locked behavior
Path traversal / key injection	Object keys are server-generated from trusted IDs and validated fixed tokens; filename never becomes path authority.
MIME spoof	Signature + bounded decoder must agree with accepted MIME; response uses validated MIME + nosniff.
Decompression bomb	Hard pixel/dimension/byte/frame limits; decoder runs with resource limits.
Active content	SVG and arbitrary documents rejected in v1; no raw HTML/script image format path.

Threat/control	DOC-17 locked behavior
EXIF privacy	Canonical original may preserve accepted input metadata but is private/protected only. EVERY anonymously public derivative, including native_webp PublicMedia.url baseline, strips GPS/EXIF/comments/nonessential application metadata.
Origin exposure	Buckets private; no object_key/bucket/checksum/private credential in public/Admin API body.
Tenant confusion	site_id/site_key/media_id verified together; wrong-site outward non-disclosure.
Unauthorized cache	Preview/Admin no-store; public cache only after public eligibility.
Deleted/private stale URL	Origin denies; CDN purge required on revocation/delete.
Transform DoS	No public arbitrary resize/crop/quality transform API; fixed catalog only.

29. Failure, Retry and Recovery Behavior

ID	Condition	Required outcome
FAIL17-01	Request fails auth/site validation	No storage write; non-disclose per upstream.
FAIL17-02	Upload rejected type/size/decode	422; no DB row/original.
FAIL17-03	Origin write fails	503; no DB row. Cleanup partial staging.
FAIL17-04	DB insert fails after object write	No MediaResponse success; object is orphan cleanup candidate, never public.
FAIL17-05	native_webp baseline generation fails for new upload	Media may remain ACTIVE for Admin recovery, but ordinary PublicMedia is omitted/MISSING until sanitized baseline succeeds; private original is never exposed.
FAIL17-06	One non-baseline derivative fails	Do not insert that variant row; retry same site/media/key; native_webp/other valid sanitized variants remain.
FAIL17-07	AVIF generation repeatedly fails	Sanitized WebP delivery continues; AVIF omitted until repaired. No fabricated URL.
FAIL17-08	Replace validation/origin/native_webp generation fails before DB switch	Old generation stays current and publicly safe; no consumer switch.
FAIL17-09	Replace commits; remaining derivatives delayed	New generation already has verified native_webp baseline; variants may be partial, old variants are not returned.
FAIL17-10	Delete finds current usage	Reject; no tombstone or blob purge.
FAIL17-11	CDN purge temporarily fails	Origin/public-eligibility revocation still denies cache misses; retry purge via later worker/ops path and treat stale-edge risk as high priority.
FAIL17-12	Wrong-site CDN locator	Neutral not-found/no-store; do not reveal owning site or valid key.

29.1 Recovery invariants

- Recovery never changes media.site_id.
- A retry never trusts client object_key or a previous worker tenant guess.
- A variant can be regenerated from the current original; media_variants is derived state.
- A failed replace never falls back to another media_id or another site image.
- A tombstoned media record is not reactivated by merely finding its blob. Any future restore capability would require upstream product/API authority and is not in v1.

30. Operational Signals and Later-Worker Handoff

DOC-17 defines what media work must happen and the identity it must retain; PDF-18 later defines queue names, job schema, retry timing, dead-letter handling and cache worker implementation. DOC-17 does not create a jobs table or worker API.

Required signal/event meaning	Minimum semantic payload	Later owner
Derivative requested	site_id, media_id, current generation identity, requested fixed variant_key set	PDF-18 exact job payload/retry.
Derivative completed/failed	site_id, media_id, generation, variant_key, outcome	PDF-18 observability.
Media replaced	site_id, media_id, old/new generation identity; public cache impact	PDF-18 cache worker implementation.
Media deleted/public eligibility revoked	site_id, media_id; purge/rebuild required	PDF-18 + later security/ops.
Media metadata changed	site_id, media_id; affected public projection metadata	PDF-18 PageData/cache invalidation.

WORKER TENANT RULE

A media job generated from a site-owned mutation must carry the originating site identity. The worker reloads the media under that site context before writing a variant or cache effect. media_id alone is never sufficient tenant authority.

31. Cross-Document Consistency Audit

Authority	Result	Verification
DOC-01	PASS	Shared object/CDN infrastructure is compatible with 14 isolated sites; no per-site backend stack invented.
DOC-02	PASS	Exact media permission keys/composites preserved; replace metadata now consumes media.edit conditionally through DOC-15 RC6.
DOC-03	PASS	All FC-MED-01..10 are executable: list/search/filter, metadata, usage indicators, variants, reuse, replace/delete and delivery.
DOC-04	PASS	SiteContext retained through storage, derivatives, usages and delivery; missing scope fails closed.
DOC-05	PASS	No new public Page ID/resource page created.
DOC-06	PASS	Only Media Library/Detail; variant/CDN work remains system effect.
DOC-07	PASS	Same-site featured/body/OG refs and MEDIA block semantics preserved.
DOC-08	PASS	Exactly media/media_variants/media_usages consumed; no new table/column/status; native_webp uses existing variant catalog row.
DOC-09	PASS	Public page route authority untouched; media resource delivery is not canonical page routing.
DOC-10	PASS	Media Library search/filter and Media Detail metadata/variants/usages now have RC6 transport; action permissions preserved.
DOC-11	PASS	Missing media/partial failure applies when sanitized public baseline is unavailable; no cross-site/private-original fallback.
DOC-12	PASS	Alt/caption/missing-media/responsive rules preserved; styling/crop presentation remains site-local.
DOC-13	PASS	PublicMedia/MediaVariant shape preserved; PublicMedia.url resolves sanitized native_webp and storage internals remain excluded.
DOC-14	PASS	Site-local UI/assets preserved; shared semantic media contract does not create site-to-site imports.
DOC-15	PASS - RC6	Three RC5 media transport gaps resolved; usage_count added; conditional media.edit on replacement metadata is executable and atomic.
DOC-16	PASS	OG/image selection respected; no image sitemap or cross-site image fallback added.

AUDIT VERDICT

RC2 closes every blocking finding from the RC1 audit. DOC-15 RC6 now transports Media Library search/filter/pagination/sort, variants, title/source metadata and usage_count, while replacement metadata has explicit conditional media.edit authorization. DOC-17 RC2 also removes the unsafe original-as-public-fallback rule: canonical originals remain private and native_webp is the mandatory sanitized public baseline. No remaining blocking upstream contradiction is known within DOC-17 scope.

32. Controlled Reconciliation Registry

32.1 GAP17-API-01 - RESOLVED in DOC-15 RC6

FC-MED-03 is now executable through API15-S-060. RC6 defines same-site search, exact MIME filtering, USED/UNUSED usage filtering, page/page_size and deterministic sort with media_id tie-breaker. No new endpoint or capability was added.

32.2 GAP17-API-02 - RESOLVED in DOC-15 RC6

MediaResource now includes required variants[] containing current verified derivative variant_key, MIME, byte_size and optional dimensions. Media Detail can render its frozen Variants/dimensions section without exposing object_key/checksum/private origin data.

32.3 GAP17-API-03 - RESOLVED in DOC-15 RC6

MediaUploadInput, MediaUpdateInput and MediaResource now carry title and source_url. Metadata PATCH remains media.view + media.edit. API15-S-064 replacement keeps media.replace for binary authority and conditionally requires media.edit when any metadata field is supplied; failure is atomic.

32.4 GAP17-API-04 - Usage indicator transport settled in DOC-15 RC6

MediaResource now requires nonnegative usage_count, so Media Library can display the FC-MED-01 usage indicator without N+1 usage-detail calls. The dedicated usage endpoint remains the detailed reference list and does not grant target-resource read authority.

32.5 GAP17-PRIV-01 - RESOLVED in DOC-17 RC2

RC1 preserved canonical original bytes but also used them as anonymous public fallback, which could expose embedded GPS/EXIF/comment metadata. RC2 keeps canonical original bytes private/protected only and reuses the already-defined native_webp variant as the mandatory metadata-sanitized full-dimension PublicMedia.url baseline. This adds no database field, variant key, Page ID or API operation.

32.6 Focal point remains a deliberate fixed rule, not a hidden gap

The finalized DOC-17 scope requires focal-point rules but upstream does not freeze an editable focal-point field. RC2 therefore preserves center focal behavior only. A future editor-controlled focal point requires controlled DOC-08/DOC-15 field and transport change; it is not assumed here.

33. Traceability Matrix

Trace ID	Source	Locked requirement	DOC-17 home
TR17-001	DOC-01 topology	Shared storage/CDN with site-isolated logical ownership	Sections 3-4
TR17-002	DOC-02 media permissions	Exact view/upload/edit/replace/delete authorization	Sections 7,17,20-21,25
TR17-003	DOC-03 FC-MED-01..10	Media capability implementation map	Sections 6-25
TR17-004	DOC-04 SiteContext	site_id carried through objects/variants/usages/jobs/cache	Sections 4,8,18,24,30
TR17-005	DOC-06 ADM-S-MEDIA-01	List/search/upload/select/reuse without new page	Sections 7,19,25
TR17-006	DOC-06/10 ADM-S-MEDIA-02	Metadata/variants/usages/replace/delete	Sections 12,17-21,25

Trace ID	Source	Locked requirement	DOC-17 home
TR17-007	DOC-07 featured_media_id	Same-site featured media delivery	Sections 18,22,26
TR17-008	DOC-07 MEDIA block	Same-site body media + alt/caption override	Sections 17,18,26
TR17-009	DOC-07 og_image_media_id	Same-site OG media source	Sections 18,27
TR17-010	DOC-08 DB-TBL-013	Original identity/storage metadata/tombstone	Sections 5,8-10,21
TR17-011	DOC-08 DB-TBL-014	Derived variant persistence	Sections 12-14
TR17-012	DOC-08 DB-TBL-023	Derived usage index	Section 18
TR17-013	DOC-08 RLS/same-site FK	Cross-site storage/use protection	Section 4
TR17-014	DOC-10 FC-MED-09	System variants effect	Sections 12-14,30
TR17-015	DOC-10 FC-MED-10	Public delivery integration	Sections 22-24
TR17-016	DOC-11 MED11-01..06	Missing media and delegated CDN/focal/variant behavior	Sections 11,16,26,29
TR17-017	DOC-12 image accessibility	Contextual alt/caption and safe missing behavior	Sections 16-17,26
TR17-018	DOC-13 PublicMedia	Resolved sanitized native_webp baseline URL + safe metadata	Sections 22-26
TR17-019	DOC-13 MediaVariant	variant_key/url/MIME/dimensions + order	Sections 12,26
TR17-020	DOC-14 site-local renderers	Shared semantic media, local UI/aspect treatment	Section 16
TR17-021	DOC-15 RC6 API15-S-060..066	Exact Media HTTP transport incl. search/filter/variants/usage_count/title/source consumed	Sections 17-18,20,25
TR17-022	DOC-16 OG selection	Deliver selected same-site image, no fallback redesign	Section 27
TR17-023	DOC-16 no image sitemap	Do not add image sitemap extension	Section 27
TR17-024	Later PDF-18 boundary	Events/invalidation conditions only; exact queue/cache mechanics later	Sections 24,30
TR17-025	Later PDF-19 boundary	No session/credential/secret/malware engine invention	Sections 22,28

34. Comprehensive Developer Acceptance Criteria

Acceptance ID	Family	Release-blocking assertion
QA17-001	Authority	No new permission key, capability, Admin/Public Page ID, database table/column/status or API operation is introduced by DOC-17.
QA17-002	Authority	Exactly the existing site-owned media/media_variants/media_usages persistence model remains authoritative.
QA17-003	Authority	All DOC-17 public objects preserve DOC-13 field names/presence rules and expose no storage internals.
QA17-004	Authority	DOC-16 image selection/fallback and no-image-sitemap rule remain unchanged.
QA17-005	Isolation	Every upload, variant, usage, replace, delete and delivery lookup begins with one trusted site context.
QA17-006	Isolation	Object keys are site-namespaced and never derived from a client site_id or filename.
QA17-007	Isolation	A valid SITE-02 media_id cannot be fetched/attached/replaced/deleted through SITE-01 context.
QA17-008	Isolation	Checksum equality does not create a shared cross-site media record.
QA17-009	Isolation	Missing media never falls back to another tenant by filename/title/subject.
QA17-010	Isolation	Worker/cache handoff retains source site identity; media_id alone is not tenant authority.
QA17-011	Upload	Only static JPEG, PNG, WebP and AVIF originals are accepted in v1.
QA17-012	Upload	GIF, SVG, video, audio, PDF and other unapproved types are rejected without creating a media row.
QA17-013	Upload	Actual file signature/decoder type is authoritative over extension and declared multipart MIME.
QA17-014	Upload	Upload stream is capped at 20 MiB and cannot allocate unbounded memory.
QA17-015	Upload	Decoded width/height are positive, each <=12000 px and total pixels <=80 MP.

Acceptance ID	Family	Release-blocking assertion
QA17-016	Upload	Multi-frame/animated image input is rejected.
QA17-017	Upload	Unsupported/oversize/decode failures map to the existing DOC-15 422 class rather than inventing 413/415.
QA17-018	Upload	Origin/storage failure before DB commit returns existing 503 semantics and leaves no successful MediaResource.
QA17-019	Storage	Original and variant object keys exactly follow the private site/media/generation grammar.
QA17-020	Storage	Client filename/path fragments never appear in authority portions of persisted keys.
QA17-021	Storage	Origin buckets are private and no ordinary anonymous S3 object URL is emitted.
QA17-022	Storage	media checksum/size/MIME/dimensions describe the canonical stored original generation.
QA17-023	Storage	Replacement uses a new immutable generation key rather than overwriting current original in place before validation.
QA17-024	Storage	Public URLs contain no bucket, object_key, checksum, internal site_id or private signed credential.
QA17-025	Storage	Canonical original bytes are never anonymously public; every public representation is a metadata-sanitized derivative.
QA17-026	Storage	PublicMedia.url resolves to verified current native_webp; if it is unavailable, media degrades/omits under existing missing-media behavior.
QA17-027	Processing	Uncropped variants preserve aspect ratio and are never upscaled.
QA17-028	Processing	V1 crop focal point is exactly center (0.5,0.5); no editable focal field is silently persisted.
QA17-029	Processing	thumb256 and thumb512 use the exact bounded center-crop algorithm and are omitted when they would require upscaling.
QA17-030	Processing	Only fixed allowlisted variant keys are generated; public arbitrary transform parameters are not supported.
QA17-031	Processing	Derivative processing strips nonessential/GPS/comment metadata and outputs sRGB-compatible pixels.
QA17-032	Processing	A media_variants row is not committed until its object exists and MIME/bytes/dimensions verify.
QA17-033	Processing	Variant generation retry remains in the same site/media/current generation.
QA17-034	Variants	Variant catalog contains only the fixed keys in Appendix A unless DOC-17 is versioned.
QA17-035	Variants	native_webp is the mandatory sanitized full-dimension public baseline; AVIF is preferred enhancement; canonical original is never anonymous fallback.
QA17-036	Variants	Photo/default WebP quality is 82 and AVIF quality 60; PNG/alpha uses lossless WebP and AVIF quality 70.
QA17-037	Variants	variants[] includes only current verified variants and is [] when none exist.
QA17-038	Variants	variants[] uses the frozen DOC-17 deterministic order.
QA17-039	Variants	Frontend selects by variant_key/MIME and does not assume array position means best source.
QA17-040	Variants	Frontend preference is AVIF -> WebP -> PublicMedia.url for equivalent geometry, and PublicMedia.url is sanitized native_webp.
QA17-041	Metadata	Media Library alt/title/caption/credit/source_url edits require media.view + media.edit.
QA17-042	Metadata	Post-local MEDIA alt_override/caption remain owned by Post edit permissions, not media.edit.
QA17-043	Metadata	Media Library blank/whitespace alt normalizes to absent; contextual empty-alt remains DOC-12 renderer behavior.
QA17-044	Metadata	Body MEDIA uses alt_override before media alt_text; card/avatar redundancy may intentionally produce empty alt per DOC-12.
QA17-045	Metadata	Metadata does not change site ownership or binary object identity except updated_at/cache revision.
QA17-046	Metadata	DOC-15 RC6 transports title and source_url in upload/update/resource schemas; source_url remains excluded from ordinary PublicMedia.
QA17-047	Usage	media_usages is derived from current authoritative Post/Author/SiteSetting references and is not the referential source of truth.

Acceptance ID	Family	Release-blocking assertion
QA17-048	Usage	POST_FEATURED/POST_BODY/POST_OG/AUTHOR_PHOTO/SITE_LOGO/SITE_FAVICON/ SITE_DEFAULT_SOCIAL usages remain same-site.
QA17-049	Usage	Delete preflight rechecks authoritative current references rather than trusting only a possibly stale usage index.
QA17-050	Usage	Usage endpoint does not grant permission to open/mutate referenced resources.
QA17-051	Usage	Usage labels do not leak protected target content when caller lacks target read permission.
QA17-052	Usage	Reuse requires media.view plus target mutation permission and revalidates same-site ownership on commit.
QA17-053	Usage	Media Library search/filter/pagination/sort is executed server-side inside the one active SiteContext with RC6 limits/defaults.
QA17-054	Usage	MediaResource.usage_count is authoritative for list/detail usage indicator display and avoids N+1 detailed usage requests.
QA17-055	Usage	MediaResource.variants exposes only current verified derivative metadata and never object_key/checksum.
QA17-056	Mutation safety	Replace preserves media_id and site_id and therefore preserves valid same-site references.
QA17-057	Mutation safety	Replace validates new binary fully before switching the current database object pointer.
QA17-058	Mutation safety	Failed replace before commit leaves old original/variants authoritative.
QA17-059	Mutation safety	Successful replace withdraws old variant rows before new-generation variants can be projected.
QA17-060	Mutation safety	Replace metadata fields alt_text/title/caption/credit/source_url update only when supplied and require additional media.edit; omitted values preserve current metadata; failure is atomic.
QA17-061	Mutation safety	Concurrent replace/delete cannot interleave into a DB/object mismatch even though DOC-15 exposes no expected_version.
QA17-062	Mutation safety	Delete is blocked by any current authoritative Post/Author/SiteSetting media reference.
QA17-063	Mutation safety	Historical revision-only media reference does not by itself block tombstone; restore must revalidate ACTIVE media.
QA17-064	Mutation safety	Successful delete sets existing DOC-08 DELETED tombstone semantics and revokes delivery; no new delete status is created.
QA17-065	Mutation safety	Physical purge is not treated as equivalent to logical delete and follows later approved retention/security policy.
QA17-066	Delivery/cache	Ordinary public delivery requires ACTIVE same-site media plus current public-eligibility predicate.
QA17-067	Delivery/cache	Draft/private-only media is not exposed through anonymous public CDN solely because it is ACTIVE.
QA17-068	Delivery/cache	Preview/Admin media delivery is protected and no-store and never becomes a shared public cache entry.
QA17-069	Delivery/cache	CDN lookup validates site_key/media_id ownership; wrong-site outcome is neutral and no-store.
QA17-070	Delivery/cache	Public URLs are fully resolved server-side; site frontends do not concatenate object-store keys.
QA17-071	Delivery/cache	Public cache key includes site namespace and v revision; cache is never keyed by media_id alone.
QA17-072	Delivery/cache	Public media may use 1-year immutable caching only with the versioned delivery URL and revocation/purge controls.
QA17-073	Delivery/cache	Replace/delete/public-eligibility loss triggers media CDN purge/revocation.
QA17-074	Delivery/cache	Deleted media old URL is not accepted as proof of active state; origin delivery denies it.
QA17-075	Delivery/cache	Metadata-only edits refresh PageData/cache metadata even when binary CDN purge is unnecessary.
QA17-076	Release gate	No failure path returns a fabricated variant URL or raw object key as fallback.
QA17-077	Release gate	Partial AVIF failure never removes a valid sanitized WebP source.
QA17-078	Release gate	Failure of non-baseline WebP derivatives is retryable degradation; failure of native_webp prevents anonymous PublicMedia until repaired and never exposes the original.
QA17-079	Release gate	An orphan object after DB failure is cleanup-only and cannot be projected as Media.

Acceptance ID	Family	Release-blocking assertion
QA17-080	Release gate	GAP17-API-01 is verified resolved by DOC-15 RC6 Media Library query/pagination/sort transport.
QA17-081	Release gate	GAP17-API-02 is verified resolved by DOC-15 RC6 MediaResource variants[] transport.
QA17-082	Release gate	GAP17-API-03 is verified resolved by DOC-15 RC6 title/source_url transport and metadata permissions.
QA17-083	Release gate	GAP17-API-04 usage-indicator transport is settled by required MediaResource.usage_count; GAP17-PRIV-01 is resolved by private-original + sanitized-native_webp public baseline rules.

35. Freeze Checklist and Final Developer Lock

Freeze ID	Condition
MED17-FZ-01	Accepted MIME/size/dimension/frame validation accepted.
MED17-FZ-02	Site/object/variant key grammar accepted.
MED17-FZ-03	Original integrity and immutability rules accepted.
MED17-FZ-04	Center focal/crop + no-upscale rule accepted.
MED17-FZ-05	Fixed WebP/AVIF responsive + thumbnail catalog accepted.
MED17-FZ-06	PublicMedia/MediaVariant resolution and ordering accepted.
MED17-FZ-07	Metadata/contextual alt behavior accepted.
MED17-FZ-08	Usage/reuse/delete safety accepted.
MED17-FZ-09	Replacement atomicity and reference preservation accepted.
MED17-FZ-10	Public/private/Preview/Admin delivery boundaries accepted.
MED17-FZ-11	CDN caching/purge/revocation rules accepted.
MED17-FZ-12	Cross-site isolation tests accepted.
MED17-FZ-13	Failure/recovery behaviors accepted.
MED17-FZ-14	GAP17-API-01 search/filter/pagination/sort resolved in DOC-15 RC6.
MED17-FZ-15	GAP17-API-02 variants transport resolved in DOC-15 RC6.
MED17-FZ-16	GAP17-API-03 title/source metadata resolved in DOC-15 RC6.
MED17-FZ-17	Usage indicator transport resolved through MediaResource.usage_count.
MED17-FZ-18	Canonical original is private; sanitized native_webp is mandatory public baseline.
MED17-FZ-19	Replacement metadata requires conditional media.edit and native_webp is verified before generation switch.
MED17-FZ-20	No later-document queue/security/deployment/visual scope has been pulled into DOC-17.

FINAL LOCK

RC2 reconciliation is complete. Implementation is conformant only when the private canonical original, sanitized native_webp public baseline, derived rows, usage index, RC6 Admin transport, site isolation, delivery URLs, variant ordering, replacement/delete semantics and CDN revocation behavior all match this specification. A faster pipeline is not equivalent if it weakens tenant identity, exposes original metadata/origin internals or changes the frozen variant/metadata/public contracts.

Appendix A - Canonical Variant Registry

#	variant_key	Geometry	Format	Availability
1	thumb256_avif	256x256 crop	AVIF	Optional optimization; center focal crop.
2	thumb256_webp	256x256 crop	WebP	Required thumbnail baseline when source can satisfy.
3	thumb512_avif	512x512 crop	AVIF	Optional optimization.
4	thumb512_webp	512x512 crop	WebP	Required when source can satisfy.
5	w320_avif	max width 320	AVIF	Uncropped responsive.
6	w320_webp	max width 320	WebP	Uncropped responsive.
7	w640_avif	max width 640	AVIF	Uncropped responsive.
8	w640_webp	max width 640	WebP	Uncropped responsive.
9	w960_avif	max width 960	AVIF	Uncropped responsive.
10	w960_webp	max width 960	WebP	Uncropped responsive.
11	w1280_avif	max width 1280	AVIF	Uncropped responsive.
12	w1280_webp	max width 1280	WebP	Uncropped responsive.
13	w1600_avif	max width 1600	AVIF	Uncropped responsive.
14	w1600_webp	max width 1600	WebP	Uncropped responsive.
15	native_avif	original dimensions	AVIF	No crop/upscale; modern-format full-size derivative.
16	native_webp	original dimensions	WebP	No crop/upscale; baseline full-size derivative.

Appendix B - Object and Delivery Key Examples

EXAMPLE - PRIVATE KEYS VS PUBLIC LOCATOR

```
Example owner: site_id=10000000-0000-4000-8000-000000000005, site_key=SITE-05
Example media_id=550e8400-e29b-41d4-a716-446655440000

private original object:
sites/10000000-0000-4000-8000-000000000005/media/550e8400-e29b-41d4-a716-446655440000/g/<sha256>/original.jpg

private derivative object:
sites/10000000-0000-4000-8000-000000000005/media/550e8400-e29b-41d4-a716-446655440000/g/<sha256>/variants/
w640_webp.webp

public resolved locator (illustrative configured CDN origin):
https://<media-cdn>/m/SITE-05/550e8400-e29b-41d4-a716-446655440000/w640_webp.webp?v=<updated_at_epoch_ms>
```

IMPORTANT

The object-key examples are normative in grammar; the CDN hostname is deployment configuration. The literal checksum is never copied into the public URL. The example does not authorize direct object-store access.

Appendix C - Canonical Usage Mapping

usage_kind	Owning current field/document	Delete blocker?	Public eligibility may arise?
POST_FEATURED	posts.featured_media_id	YES	YES when Post PUBLISHED/public.
POST_BODY	ArticleBody MEDIA block	YES	YES when Post PUBLISHED/public.

usage_kind	Owning current field/document	Delete blocker?	Public eligibility may arise?
POST_OG	posts.og_image_media_id	YES	YES when selected by public SEO/OG.
AUTHOR_PHOTO	author_site_profiles.photo_media_id	YES	YES when Author public-eligible.
SITE_LOGO	site_settings.logo_media_id	YES	YES on public shell/structured data when used.
SITE_FAVICON	site_settings.favicon_media_id	YES	YES when emitted for ACTIVE site.
SITE_DEFAULT_SOCIAL	site_settings.default_social_image_media_id	YES	YES when DOC-16 fallback selects it.

Appendix D - Failure Matrix

Failure ID	Before/after durable switch	Visible outcome	Authoritative data after failure
FAIL17-01	Before commit	No storage write; non-disclose per upstream.	No new media/current old generation
FAIL17-02	Before commit	422; no DB row/original.	No new media/current old generation
FAIL17-03	Before commit	503; no DB row. Cleanup partial staging.	No new media/current old generation
FAIL17-04	Before commit	No MediaResponse success; object is orphan cleanup candidate, never public.	No new media/current old generation
FAIL17-05	After/independent	Media may remain ACTIVE for Admin recovery, but ordinary PublicMedia is omitted/MISSING until sanitized baseline succeeds; private original is never exposed.	Current DB row + only verified objects/variants
FAIL17-06	After/independent	Do not insert that variant row; retry same site/media/key; native_webp/other valid sanitized variants remain.	Current DB row + only verified objects/variants
FAIL17-07	After/independent	Sanitized WebP delivery continues; AVIF omitted until repaired. No fabricated URL.	Current DB row + only verified objects/variants
FAIL17-08	Before commit	Old generation stays current and publicly safe; no consumer switch.	No new media/current old generation
FAIL17-09	After/independent	New generation already has verified native_webp baseline; variants may be partial, old variants are not returned.	Current DB row + only verified objects/variants
FAIL17-10	After/independent	Reject; no tombstone or blob purge.	Current DB row + only verified objects/variants
FAIL17-11	After/independent	Origin/public-eligibility revocation still denies cache misses; retry purge via later worker/ops path and treat stale-edge risk as high priority.	Current DB row + only verified objects/variants
FAIL17-12	After/independent	Neutral not-found/no-store; do not reveal owning site or valid key.	Current DB row + only verified objects/variants

Appendix E - One-Page Developer Execution Reference

UPLOAD

SiteContext + media.view/media.upload -> stream <=20 MiB -> signature/decoder allowlist JPEG/PNG/WebP/AVIF -> <=12000x12000 and <=80MP, one frame -> private generation object -> media ACTIVE row -> derivative work. Validation is 422; storage unavailable 503.

STORAGE

Private S3-compatible keys: sites/{site_id}/media/{media_id}/g/{checksum}/... . Filename never owns a path. Origin not anonymous. Public URLs never expose site_id, object_key, bucket, checksum or private credential.

VARIANTS

Fixed catalog only. Square thumb256/thumb512 center-cropped; widths 320/640/960/1280/1600 + native; no upscale. native_webp mandatory sanitized public baseline; AVIF preferred; canonical original private only. Public variants[] contains verified current rows in fixed order.

MUTATIONS

Replace keeps media_id/site_id, stages private original + sanitized native_webp, requires media.edit if metadata is supplied, atomically switches only after baseline verification, then regenerates/purges CDN. Delete rechecks authoritative usages, blocks in-use media, tombstones DELETED, revokes delivery; physical purge is later policy.

DELIVERY

PUBLIC requires ACTIVE + same-site + current public eligibility + verified native_webp. PublicMedia.url and variants are sanitized derivatives only; original is private. ADMIN/PREVIEW are protected no-store. CDN cache keys include site namespace; replace/delete/eligibility loss purge/revoke.

RECONCILIATION

DOC-15 RC6 closes the three RC5 transport gaps and settles usage_count/replace-metadata authorization. DOC-17 RC2 closes the original-metadata privacy contradiction by keeping canonical originals private and using sanitized native_webp as public baseline. No blocking reconciliation item remains known.